

Peano 算术 (1)

定义 (G. Peano 公理) 设 N 是集合, $0 \in N$ 是给定的元素, 而 $s: N \rightarrow N$ 是映射. 若以下条件成立, 则称 $(N, 0, s)$ 满足 Peano 公理:

▷ 零非后继: 不存在 $x \in N$ 使得 $0 = s(x)$

▷ 后继元运算是单射: 对 $\forall x, y \in N$, 若 $s(x) = s(y)$, 则 $x = y$

▷ 数学归纳法原理: 设 $I \subseteq N$ 为 N 的包含 0 的子集, 而且对于 $\forall x \in N$ 都有 $x \in I \Rightarrow s(x) \in I$, 则 $I = N$.

Lemma: 设 $(N, 0, s)$ 满足 Peano 公理, $x \in N$ 而 $x \neq 0$, 则必存在 $y \in N$ 使得 $x \neq y$ 而 $x = s(y)$.

Proof: 假设对 $\forall \alpha \in N$, 有: $x = \alpha$ 或 $x \neq s(\alpha)$.

考虑集合 $N \setminus \{x\}$. $\because 0 \in N$ 且 $0 \neq x \therefore 0 \in N \setminus \{x\} \therefore N \setminus \{x\} \neq \emptyset$

$\because N \setminus \{x\} \subseteq N \therefore N \setminus \{x\}$ 是 N 的包含 0 的子集.

对 $\forall \lambda \in N$. 若 $\lambda \in N \setminus \{x\}$, 则有: $\lambda \in N$ 且 $\lambda \neq x$.

$\because \lambda \in N \therefore x = \lambda$ 或 $x \neq s(\lambda) \therefore x \neq \lambda \therefore x \neq s(\lambda)$

$\because \lambda \in N, s: N \rightarrow N$ 是映射 $\therefore s(\lambda) \in N \therefore s(\lambda) \in N \setminus \{x\}$.

$\therefore N \setminus \{x\} = N \therefore x \in N = N \setminus \{x\} \therefore x \neq x$. 矛盾.

$\therefore \exists y \in N$, 使得 $x \neq y$ 且 ~~$x \neq s(y)$~~
 $x = s(y)$



定理 (递归原理) 设 $(N, 0, s)$ 满足 Peano 公理, Y 是任意集合, $t: Y \rightarrow Y$ 是映射, 而 $y_0 \in Y$. 此时存在唯一的映射 $f: N \rightarrow Y$ 使得 $f(0) = y_0$ 而且对 $\forall x \in N$, 有 $f(s(x)) = t(f(x))$.

Proof: 先证明唯一性. 假设存在两个映射 $f_1: N \rightarrow Y$, $f_2: N \rightarrow Y$, 满足

$$f_1(0) = y_0 \text{ 且对 } \forall x \in N, \text{ 有 } f_1(s(x)) = t(f_1(x))$$

$$f_2(0) = y_0 \text{ 且对 } \forall x \in N, \text{ 有 } f_2(s(x)) = t(f_2(x))$$

$$\text{令 } \Delta := \{x \in N : f_1(x) = f_2(x)\}. \quad \therefore \Delta \subseteq N$$

$$\therefore 0 \in N, \text{ 且有 } f_1(0) = y_0 = f_2(0) \quad \therefore 0 \in \Delta \quad \therefore \Delta \text{ 是 } N \text{ 的包含 } 0 \text{ 的子集.}$$

$$\text{对 } \forall \alpha \in N. \text{ 若 } \alpha \in \Delta, \text{ 则有: } f_1(\alpha) = f_2(\alpha)$$

$$\therefore s: N \rightarrow N \text{ 是映射} \quad \therefore s(\alpha) \in N.$$

$$\therefore f_1(s(\alpha)) = \underset{\substack{\downarrow \\ t \text{ 是映射}}}{t(f_1(\alpha))} = t(f_2(\alpha)) = f_2(s(\alpha)) \quad \therefore s(\alpha) \in \Delta$$

$$\therefore \Delta = N. \quad \therefore \text{对 } \forall x \in N, \text{ 有: } x \in N = \Delta \quad \therefore f_1(x) = f_2(x) \quad \therefore f_1 = f_2$$

$\therefore$ 唯一性得证.

$$\text{下证存在性. 令 } \mathcal{U} := \{ \Gamma \subseteq N \times Y \mid (0, y_0) \in \Gamma, \text{ 且对 } \forall (x, y) \in \Gamma, \text{ 有 } (s(x), t(y)) \in \Gamma \}.$$

$$\text{显然有 } N \times Y \subseteq N \times Y. \quad \therefore 0 \in N \text{ 且 } y_0 \in Y \quad \therefore (0, y_0) \in N \times Y.$$

$$\text{对 } \forall (x, y) \in N \times Y, \text{ 有: } x \in N \text{ 且 } y \in Y. \quad \therefore s: N \rightarrow N \text{ 和 } t: Y \rightarrow Y \text{ 都是映射}$$

$$\therefore s(x) \in N \text{ 且 } t(y) \in Y. \quad \therefore (s(x), t(y)) \in N \times Y. \quad \therefore N \times Y \in \mathcal{U}.$$

$$\therefore \mathcal{U} \neq \emptyset. \quad \text{令 } \Gamma_{\min} := \bigcap_{\Gamma \in \mathcal{U}} \Gamma.$$

$$\Gamma_{\min} = \bigcap_{\Gamma \in \mathcal{U}} \Gamma \subseteq N \times Y.$$

$$\therefore \text{对 } \forall \Gamma \in \mathcal{U}, \text{ 有 } (0, y_0) \in \Gamma \quad \therefore (0, y_0) \in \bigcap_{\Gamma \in \mathcal{U}} \Gamma = \Gamma_{\min}.$$

$$\text{对 } \forall (x, y) \in \Gamma_{\min}, \text{ 有: } (x, y) \in \Gamma_{\min} = \bigcap_{\Gamma \in \mathcal{U}} \Gamma$$

$$\therefore \text{对 } \forall \Gamma \in \mathcal{U}, \text{ 有 } (x, y) \in \Gamma. \quad \therefore (x, y) \in \Gamma \quad \therefore (s(x), t(y)) \in \Gamma$$

$$\therefore (s(x), t(y)) \in \bigcap_{\Gamma \in \mathcal{U}} \Gamma = \Gamma_{\min} \quad \therefore \Gamma_{\min} \in \mathcal{U}.$$

对 $\forall x \in N$, 定义如下的命题:

$$P(x): \text{对 } \forall y, y' \in Y, \text{ 若 } (x, y), (x, y') \in \Gamma_{\min}, \text{ 则有: } y = y'.$$

$$Q(x): \exists y \in Y, \text{ 使得: } (x, y) \in \Gamma_{\min}$$

$$\text{令 } I := \{x \in N : P(x) \wedge Q(x)\}. \quad \therefore I \subseteq N.$$

$$\text{对于 } 0 \in N. \quad \therefore \exists y_0 \in Y, \text{ 使得 } (0, y_0) \in \Gamma_{\min} \quad \therefore Q(0) \text{ 成立.}$$

$$\text{令 } \Gamma_1 := \{(0, y_0)\} \cup ((N \setminus \{0\}) \times Y).$$

$$\therefore (0, y_0) \in N \times Y \quad \therefore \{(0, y_0)\} \subseteq N \times Y. \quad \therefore (N \setminus \{0\}) \times Y \subseteq N \times Y$$

$$\therefore \Gamma_1 = \{(0, y_0)\} \cup ((N \setminus \{0\}) \times Y) \subseteq N \times Y$$

$$\text{显然有 } (0, y_0) \in \{(0, y_0)\} \subseteq \{(0, y_0)\} \cup ((N \setminus \{0\}) \times Y) = \Gamma_1$$

$$\text{对 } \forall (x, y) \in \Gamma_1 = \{(0, y_0)\} \cup ((N \setminus \{0\}) \times Y), \text{ 有: } (x, y) \in \{(0, y_0)\} \text{ 或 } (x, y) \in (N \setminus \{0\}) \times Y$$

$$\text{若 } (x, y) \in \{(0, y_0)\}, \text{ 则有: } (x, y) = (0, y_0) \quad \therefore x = 0 \text{ 且 } y = y_0.$$

$$\therefore s(x) = s(0) \neq 0, \quad \therefore s(x) \in N \setminus \{0\}. \quad \therefore t(y) = t(y_0) \in Y$$

$$\therefore (s(x), t(y)) \in (N \setminus \{0\}) \times Y \subseteq \Gamma_1$$

若 $(x, y) \in (N \setminus \{0\}) \times Y$, 则有: $x \in N \setminus \{0\}$ 且 $y \in Y$.

$\therefore s(x) \neq 0 \quad \therefore s(x) \in N \setminus \{0\} \quad \therefore y \in Y \quad \therefore t(y) \in Y$

$\therefore (s(x), t(y)) \in (N \setminus \{0\}) \times Y \subseteq \Gamma_1$

$\therefore \Gamma_1 \in \mathcal{U}$

对于 $0 \in N$. 对 $\forall y, y' \in Y$, 若 $(0, y), (0, y') \in \Gamma_{\min}$, 则有:

$\therefore (0, y) \in \Gamma_{\min} = \bigcap_{\Gamma \in \mathcal{U}} \Gamma \quad \therefore (0, y) \in \Gamma_1 \quad \therefore 0 \notin N \setminus \{0\}$

$\therefore (0, y) \in \{(0, y_0)\} \quad \therefore y = y_0$

$\therefore (0, y') \in \Gamma_{\min} = \bigcap_{\Gamma \in \mathcal{U}} \Gamma \quad \therefore (0, y') \in \Gamma_1 \quad \therefore 0 \notin N \setminus \{0\}$

$\therefore (0, y') \in \{(0, y_0)\} \quad \therefore y' = y_0 \quad \therefore y = y_0 = y' \quad \therefore P(0) \text{ 成立.}$

$\therefore 0 \in N$ 且 $P(0)$ 成立 且 $Q(0)$ 成立 $\therefore 0 \in I$

$\therefore I$ 是 N 的包含 0 的子集.

对 $\forall x \in N$. 若 $x \in I$, 则有:

$\therefore x \in I \quad \therefore x \in N$ 且 $P(x)$ 成立 且 $Q(x)$ 成立.

$\therefore x \in N$, $s: N \rightarrow N$ 是映射 $\therefore s(x) \in N$.

$\therefore x \in N$ 且 $Q(x)$ 成立 $\therefore \exists \theta \in Y$, 使得 $(x, \theta) \in \Gamma_{\min}$

$\therefore \Gamma_{\min} \in \mathcal{U} \quad \therefore$ 对于 $(x, \theta) \in \Gamma_{\min}$, 有: $(s(x), t(\theta)) \in \Gamma_{\min}$

$\therefore \theta \in Y$, $t: Y \rightarrow Y$ 是映射 $\therefore t(\theta) \in Y$.

$\therefore$ 对于 $s(x) \in N$, $\exists t(\theta) \in Y$, 使得 $(s(x), t(\theta)) \in \Gamma_{\min} \quad \therefore Q(s(x))$ 成立.

对于 $s(x) \in N$, 对 $\forall \varphi, \varphi' \in Y$. 若 $(s(x), \varphi), (s(x), \varphi') \in \Gamma_{\min}$, 则有:

假设 $\varphi \neq t(\theta)$, 则有:

$$\because (s(x), \varphi) \in \Gamma_{\min} \quad \therefore \text{令 } \Gamma_1 = \Gamma_{\min} \setminus \{(s(x), \varphi)\} \quad \therefore \Gamma_1 \subseteq \Gamma_{\min} \subseteq N \times Y$$

$$\because (0, y_0) \in \Gamma_{\min}, \text{ 又 } (0, y_0) \neq (s(x), \varphi) \quad \therefore (0, y_0) \in \Gamma_1$$

$$\text{对 } \forall (\gamma, d') \in \Gamma_1, \text{ 有: } (\gamma, d') \in \Gamma_{\min} \text{ 且 } (\gamma, d') \neq (s(x), \varphi)$$

$$\because \Gamma_{\min} \in \mathcal{U}, (\gamma, d') \in \Gamma_{\min} \quad \therefore (s(\gamma), t(d')) \in \Gamma_{\min}$$

$$\text{假设 } (s(\gamma), t(d')) = (s(x), \varphi), \text{ 则有: } s(\gamma) = s(x) \text{ 且 } t(d') = \varphi.$$

$$\because s: N \rightarrow N \text{ 是单射} \quad \therefore x = \gamma \quad \therefore (x, d') \in \Gamma_{\min}$$

$$\because (\gamma, d') \in \Gamma_1 \subseteq \Gamma_{\min} \subseteq N \times Y \quad \therefore \gamma \in N \text{ 且 } d' \in Y.$$

$$\because x \in N, \theta \in Y, d' \in Y, (x, \theta) \in \Gamma_{\min}, (x, d') \in \Gamma_{\min}, P(x) \text{ 成立} \quad \therefore \theta = d'$$

$$\therefore \varphi = t(d') = t(\theta), \text{ 这与 } \varphi \neq t(\theta) \text{ 矛盾.} \quad \therefore (s(\gamma), t(d')) \neq (s(x), \varphi)$$

$$\therefore (s(\gamma), t(d')) \in \Gamma_1 \quad \therefore \Gamma_1 \in \mathcal{U}$$

$$\therefore \Gamma_{\min} = \bigcap_{\Gamma \in \mathcal{U}} \Gamma \subseteq \Gamma_1 \quad \therefore \Gamma_{\min} = \Gamma_1$$

$$\therefore (s(x), \varphi) \notin \Gamma_1 = \Gamma_{\min} \text{ 矛盾.} \quad \therefore \varphi = t(\theta).$$

假设 $\varphi' \neq t(\theta)$, 则有:

$$\because (s(x), \varphi') \in \Gamma_{\min} \quad \therefore \text{令 } \Gamma_2 = \Gamma_{\min} \setminus \{(s(x), \varphi')\} \quad \therefore \Gamma_2 \subseteq \Gamma_{\min} \subseteq N \times Y$$

$$\because (0, y_0) \in \Gamma_{\min}, \text{ 又 } (0, y_0) \neq (s(x), \varphi') \quad \therefore (0, y_0) \in \Gamma_2$$

$$\text{对 } \forall (\varepsilon, \zeta) \in \Gamma_2, \text{ 有: } (\varepsilon, \zeta) \in \Gamma_{\min} \text{ 且 } (\varepsilon, \zeta) \neq (s(x), \varphi')$$

$$\because \Gamma_{\min} \in \mathcal{U}, (\varepsilon, \zeta) \in \Gamma_{\min} \quad \therefore (s(\varepsilon), t(\zeta)) \in \Gamma_{\min}$$

$$\text{假设 } (s(\varepsilon), t(\zeta)) = (s(x), \varphi'), \text{ 则有: } s(\varepsilon) = s(x) \text{ 且 } t(\zeta) = \varphi'$$

$$\because s: N \rightarrow N \text{ 是单射} \quad \therefore \varepsilon = x \quad \therefore (x, \zeta) \in \Gamma_{\min}$$

$$\therefore (\varepsilon, \zeta) \in \Gamma_{\min} \subseteq N \times Y \quad \therefore \varepsilon \in N \text{ 且 } \zeta \in Y$$

$$\therefore x \in N, \theta \in Y, \zeta \in Y, (x, \theta) \in \Gamma_{\min}, (x, \zeta) \in \Gamma_{\min}, P(x) \text{ 成立} \quad \therefore \theta = \zeta$$

$$\therefore \varphi' = t(\zeta) = t(\theta), \text{ 矛盾.} \quad \therefore (s(\varepsilon), t(\zeta)) \neq (s(x), \varphi')$$

$$\therefore (s(\varepsilon), t(\zeta)) \in \Gamma_2 \quad \therefore \Gamma_2 \in \mathcal{U}$$

$$\therefore \Gamma_{\min} = \bigcap_{\Gamma \in \mathcal{U}} \Gamma \subseteq \Gamma_2 \quad \therefore \Gamma_{\min} = \Gamma_2$$

$$\therefore (s(x), \varphi') \notin \Gamma_2 = \Gamma_{\min} \quad \text{矛盾.} \quad \therefore \varphi' = t(\theta)$$

$$\therefore \varphi = t(\theta) = \varphi' \quad \therefore P(s(x)) \text{ 成立.}$$

$$\therefore s(x) \in N \text{ 且 } P(s(x)) \text{ 成立 且 } Q(s(x)) \text{ 成立} \quad \therefore s(x) \in I \quad \therefore I = N$$

$$\therefore \text{对 } \forall x \in N, \text{ 有: } x \in N = I \quad \therefore P(x) \text{ 和 } Q(x) \text{ 都成立.}$$

$$\therefore \exists \text{ 唯一的 } y \in Y, \text{ 使得: } (x, y) \in \Gamma_{\min}.$$

$$\text{令 } f: N \longrightarrow Y \quad \therefore f \text{ 是 } N \rightarrow Y \text{ 的映射.}$$

$$x \longmapsto y = f(x) \text{ 使得 } (x, y) \in \Gamma_{\min}$$

$$\therefore s \text{ 是 } N \rightarrow N \text{ 的映射} \quad \therefore f \circ s \text{ 是 } N \rightarrow Y \text{ 的映射.}$$

$$\therefore t \text{ 是 } Y \rightarrow Y \text{ 的映射} \quad \therefore t \circ f \text{ 是 } N \rightarrow Y \text{ 的映射.}$$

$$\text{对 } \forall x \in N, \quad \therefore \exists \text{ 唯一的 } y \in Y, \text{ s.t. } (x, y) \in \Gamma_{\min} \quad \therefore y = f(x).$$

$$\therefore \Gamma_{\min} \in \mathcal{U}, \quad (x, y) \in \Gamma_{\min} \quad \therefore (s(x), t(y)) \in \Gamma_{\min}$$

$$\therefore s(x) \in N \quad \therefore t(y) = f(s(x))$$

$$\therefore (f \circ s)(x) = f(s(x)) = t(y) = t(f(x)) = (t \circ f)(x) \quad \therefore f \circ s = t \circ f.$$

$$\therefore (0, y_0) \in \Gamma_{\min}, \quad 0 \in N \quad \therefore y_0 = f(0)$$

$$\therefore f \text{ 是 } N \rightarrow Y \text{ 的映射, } f(0) = y_0, \quad f \circ s = t \circ f. \quad \therefore \text{存在性得证.} \quad \square$$